

Yi Hu

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EDUCATION

PhD in Biomedical Engineer

University of Alberta

Jan. 2021 – Exp. Apr. 2026

Edmonton, Canada

MSc in Control Theory and Control Engineering

Sichuan University

Sept. 2017 – Jun. 2020

Chengdu, China

B.S. in Process Equipment and Control Engineering

Hefei University of Technology

Sept. 2013 – Jun. 2017

Hefei, China

PERSONAL SUMMARY

PhD researcher specialized in embodied AI and surgical robotics. Developed VLA models and RL/IL frameworks for Franka and da Vinci surgical robots. Strong background in SLAM, kinematics/dynamics, and control theory.

PROJECT EXPERIENCE

Human-Centered Autonomy Lab

Graduate Research Assistant, University of Alberta

Jan. 2024 – Present

Edmonton, Canada

- Built a large **Vision-Language-Action (VLA)** model with **3D spatial reconstruction**. **Full-stack VLA workflow**: Collected and post-training $\pi 0.5$ model on large-scale multimodal datasets (**over 300G, 5 manipulation tasks**), deployed for real-time **RGB-D inference** on a Franka robot.
- Engineered a **real-time motion policy** for dynamic **fast manipulation** by architecting a single-step **diffusion model**; Optimized the inference engine to achieve 7-10x lower latency; Demonstrated robustness via **sim-to-real** transfer (**PyBullet** to **Franka robot**)
- Developed a **Reinforcement Learning (RL)** framework transferring **human datasets** priors to robotic surgical subtasks. Enabled optimized **patient side manipulation** and **endoscopic camera operation** from human daily demonstrations. Demonstrated robustness via **sim-to-real** transfer (**PyBullet** to **da Vinci surgical robot**)

Telerobotic and Biorobotic Systems Group

Graduate Research Assistant, University of Alberta

Jan. 2021 – Dec. 2023

Edmonton, Canada

- Built an **end-to-end** autonomous ultrasound system on a Franka robot. Fused **classical control** (admittance control) with a real-time **deep learning image processing** to regulate probe contact. Engineered data foundation that collects and **auto-labels multimodal sensor data** from **force-torque sensor** and **video grabber**.
- Developed an autonomous endoscopic camera system for the **da Vinci surgical robot** by **Imitation Learning (IL)** from noisy surgical data; Engineered a **safety-critical** and **collision-aware motion planning** framework.
- Developed a **real-time teleoperation** pipeline that transformed **free-hand movements** into **6-DoF robot** control; Integrated a lightweight **gesture classifier** trigger discrete robot actions.

Perception, Motion Control and Intelligent Robot Innovation Lab

Graduate Research Assistant, Sichuan University

Sept. 2017 – Dec. 2020

Chengdu, China

- Implemented a real-time **event-camera SLAM** pipeline for **mobile robot navigation** with **scene understanding**, enabling **6-DoF motion tracking** and **3D reconstruction** through optimization-based mapping and keyframe back-end refinement. Built a **physics-based modeling** in Unity for mobile robot navigation.
- Designed robust **nonlinear control** algorithms for **joint-space control**, accounting for **actuator saturation** and incorporated **disturbance observers**, with **system stability** guaranteed through Lyapunov analysis.
- Derived **kinematic and dynamic models** for a wheel mobile robot, enabling motion planning, control design, and trajectory tracking.

SKILLS

Programming and Frameworks: Python, MATLAB/Simulink, C++, Keras, TensorFlow, Pytorch, Linux, Ubuntu

Robotics and Tools: MuJoCo, Isaac Gym, PyBullet, ROS2, RViz, Gazebo, Git, SolidWorks, CATIA, AutoCAD, Ansys, Comsol, LaTeX

Soft Skills: Communicating cross functionally, Continuous Learning, Innovative Thinking, Event Management

Languages: Mandarin(Native), Cantonese(Native), English(Fluent)